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Personal Care





Mastering Modern Hair Care

Formulate for the Trends that Matter

Agenda

1. Global hair care market dynamics
2. Popular shampoo claims
3. Strategies for aligning product development with consumer needs

Objectives

- Market Insights
- Science-Backed Substantiation
- Empowering Formulation

Do not edit
How to change the design



Does the consumer perceive Sulfate-free as natural?

The Slido app must be installed on every computer you're presenting from

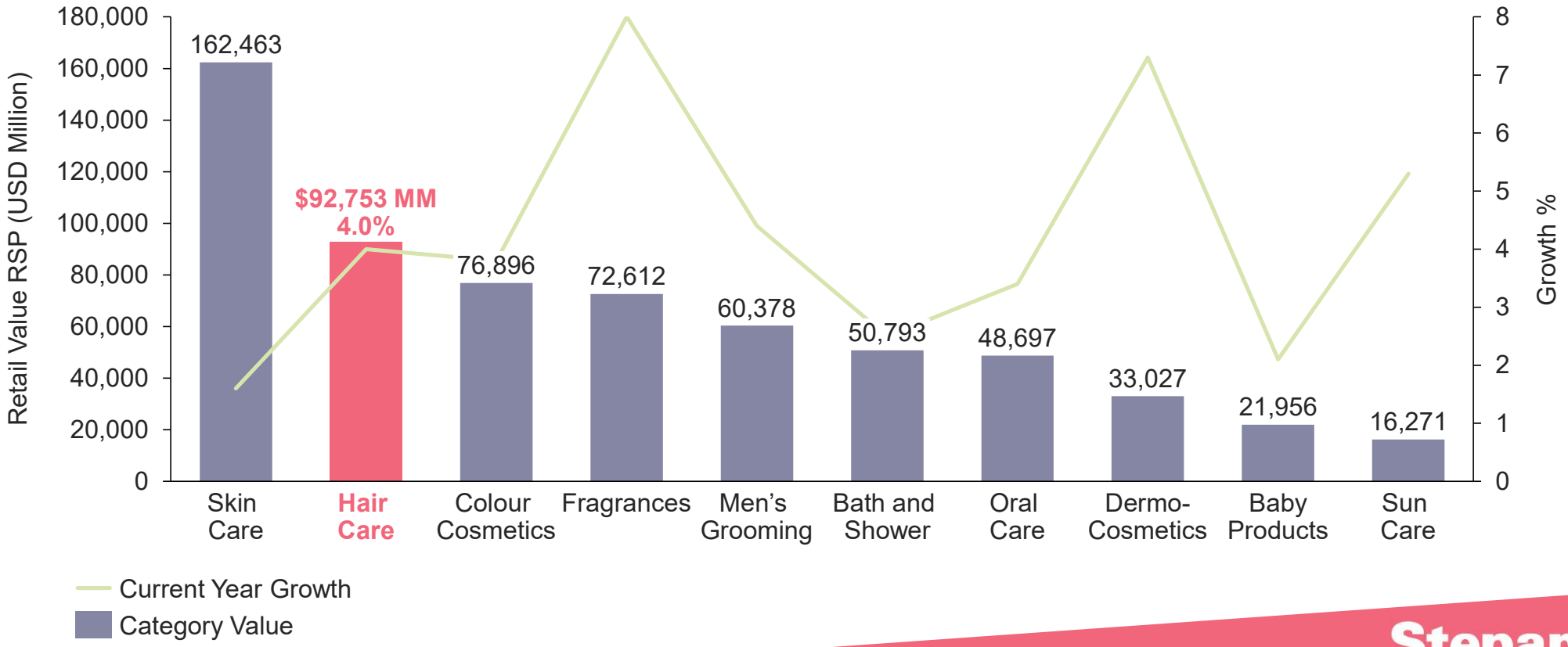
slido

Exploring the world of modern hair care

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Second to none (well, almost): global hair care is growing

Global Sales of Beauty and Personal Care by Category
Euromonitor 2025



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Trends shifting to new era of hair care: Smart & Personal



Skinification

Adopting skincare ingredients, formats, and claims



Personalization & Efficacy

Tailored solutions with data-backed claims



Premiumization

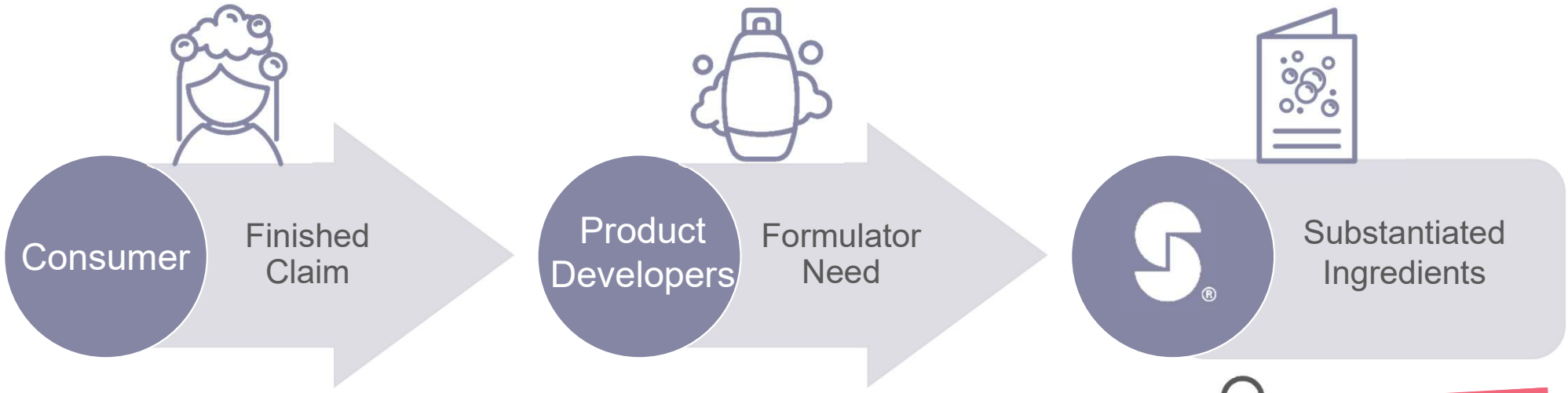
Willingness to invest in hair health and self-care

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Formulating for
Moisturizing, Anti-Static & Damaged Hair

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Formulating for top shampoo claims



Damaged Hair
Moisturizing
Anti-Static

Hair Breakage
Moisture Retention
Static Reduction

- ❑ Ether sulfate
- ❑ Ammonium sulfate

ALPHA-STEP PC-48

The clever chemistry behind ALPHA-STEP PC-48

INCI Name

Sodium Methyl 2-Sulfolaurate
(and) Disodium 2-Sulfolaurate

Appearance

Clear, light-yellow liquid

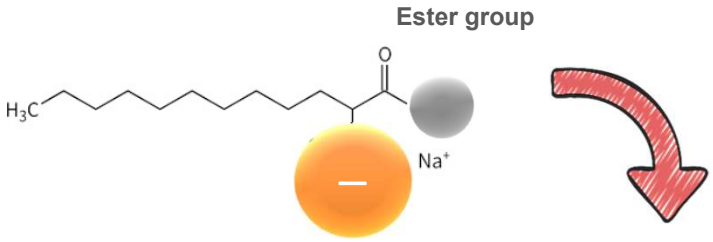
Actives

37-40%

Light, fluffy and flash foam
Rinses cleanly and easily

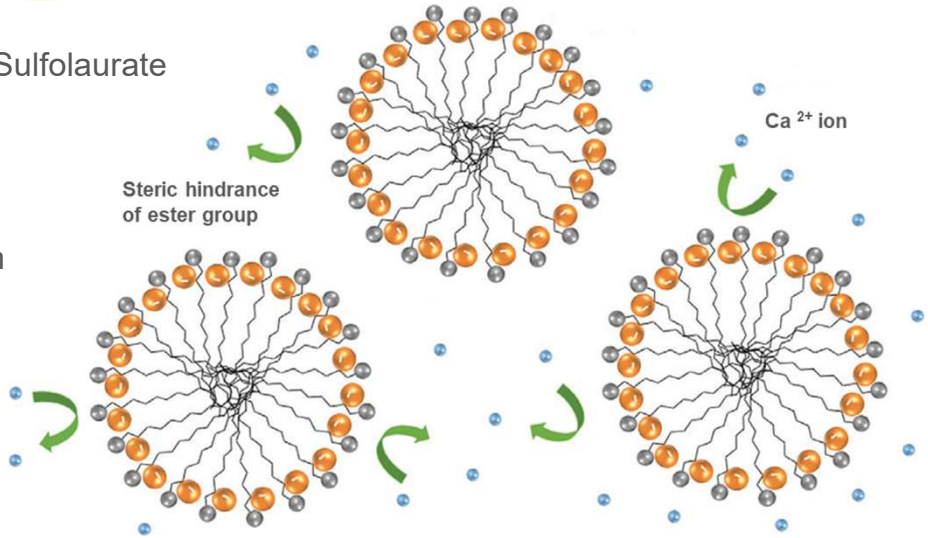
94% Bio-based
Readily biodegradable

Excellent hard water tolerance
CleanGredients® listed



Sodium Methyl 2-Sulfolaurate

Steric hindrance of sulfonic groups from α-MES under high water hardness conditions

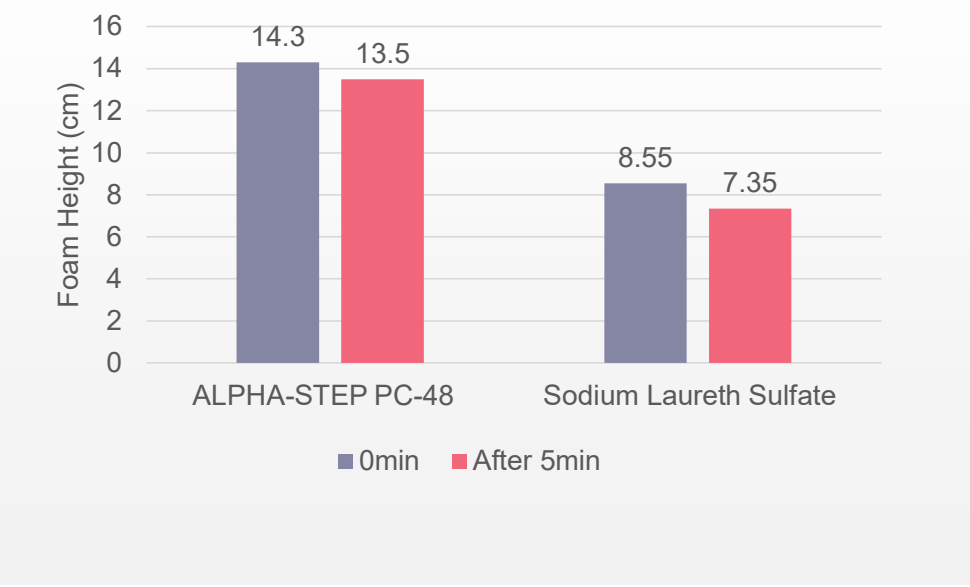


Source: Lim, Y. S., Baharudin, N. B., & Ung, Y. W. (2018). Methyl ester sulfonate: A high-performance surfactant capable of reducing builders dosage in detergents. Journal of Surfactants and Detergents. https://doi.org/10.1002/jsde.12230

Hard water tolerance made easy for formulation flexibility

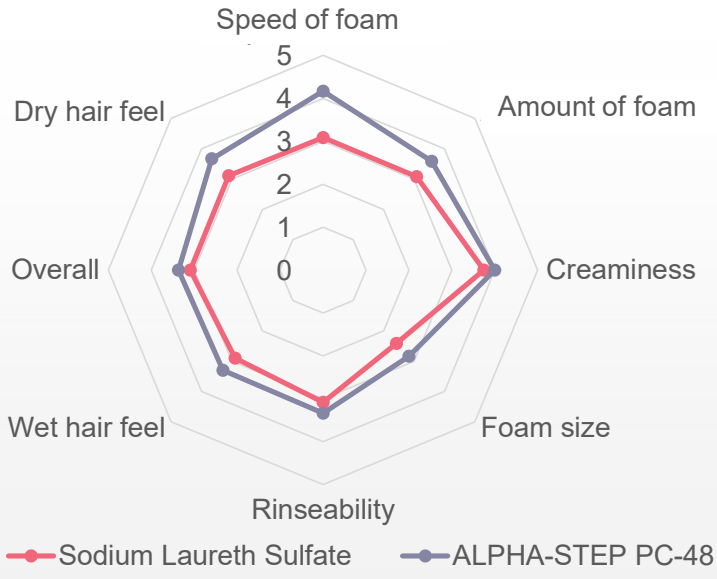
Foam Volume & Longevity

ALPHA-STEP PC-48 demonstrates significantly higher foaming compared to Sodium Laureth Sulfate in 3,000 ppm hard water.



Feel & Foam Appearance

ALPHA-STEP PC-48 improves overall feel and speed of foam compared to Sodium Laureth Sulfate in 1,000 ppm hard water.

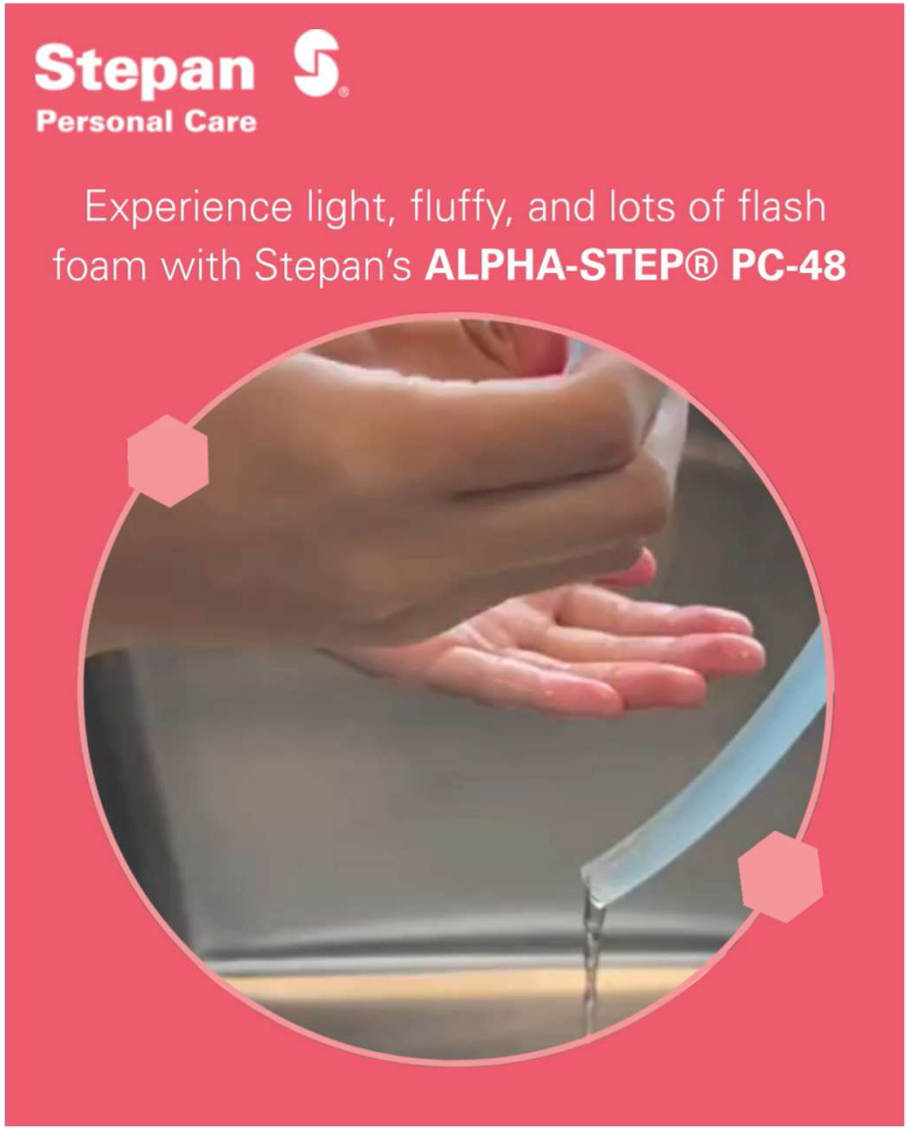


Ross-Miles Foam Tester with 0.2% active solution at 25°C was used for the foam test.
Sensory evaluation was conducted with 10 panelists on hair tresses washed using 0.5mL of sample.
Shampoo formulas: 10% active anionic surfactant, 5% active cocamidopropyl betaine.

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Pop-Up

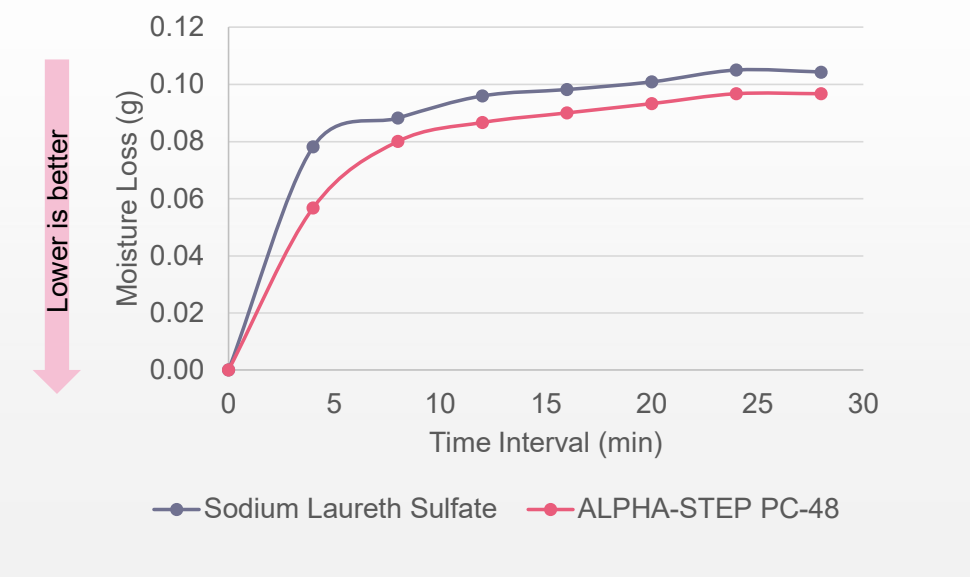
Video #1



Check off hydration and anti-static hair claims

Moisture Retention

ALPHA-STEP PC-48 demonstrates significantly less moisture loss compared to Sodium Laureth Sulfate, making it an ideal choice for formulators seeking superior hydration benefits.



Static Reduction

ALPHA-STEP PC-48 exhibits lower hair static compared to Sodium Laureth Sulfate, perfect for shampoo formulators making anti-static claims.

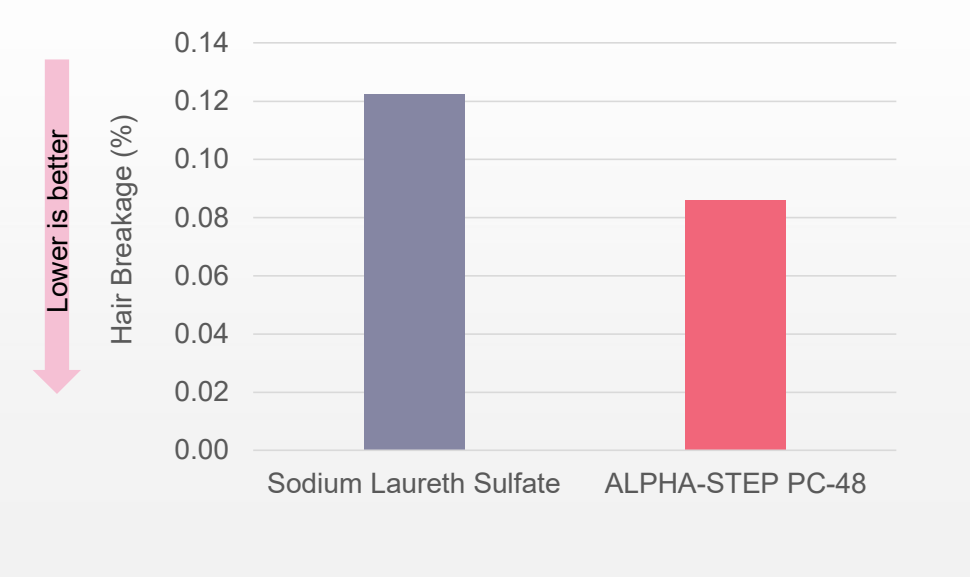


Hair tresses washed under 1,000 ppm hard water for 10 cycles (1 cycle: wetting-shampooing-rinsing-drying). Treated hair tresses dried overnight at 25 °C .
Initial weight of tress was recorded and compared to the final weight after specific time in 45 °C oven.
Shampoo formulas:10% active surfactant, 5% active Cocamidopropyl Betaine, 1% conditioning agent.

Damaged hair care routines are worry-free

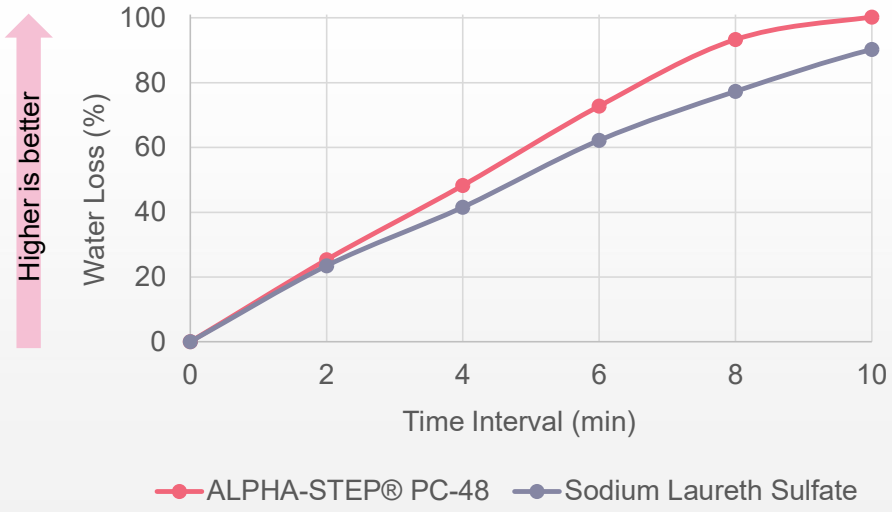
Hair Breakage by Flat Ironing

ALPHA-STEP PC-48 results in lesser hair breakage compared to Sodium Laureth Sulfate, making it a gentler alternative for shampoo formulations.



Hydrophobicity by Blow Drying

ALPHA-STEP PC-48 exhibits a faster drying rate compared to Sodium Laureth Sulfate, enhancing the overall efficiency of hair care routines.



Hair tresses washed under 1,000 ppm hard water for 10 cycles (1 cycle: wetting-shampooing-rinsing-drying). Treated hair tresses were dried overnight at 25 °C.
Hair breakage: Treated hair tresses flat ironed at 230°C for 20 times and vigorously combed for 50 times.
Hydrophobicity: Wet hair tress positioned 15 cm away from hair dryer, weight measured at 2-minute interval upon drying.
Shampoo formulas:10% active surfactant, 5% active Cocamidopropyl Betaine, 1% conditioning agent.

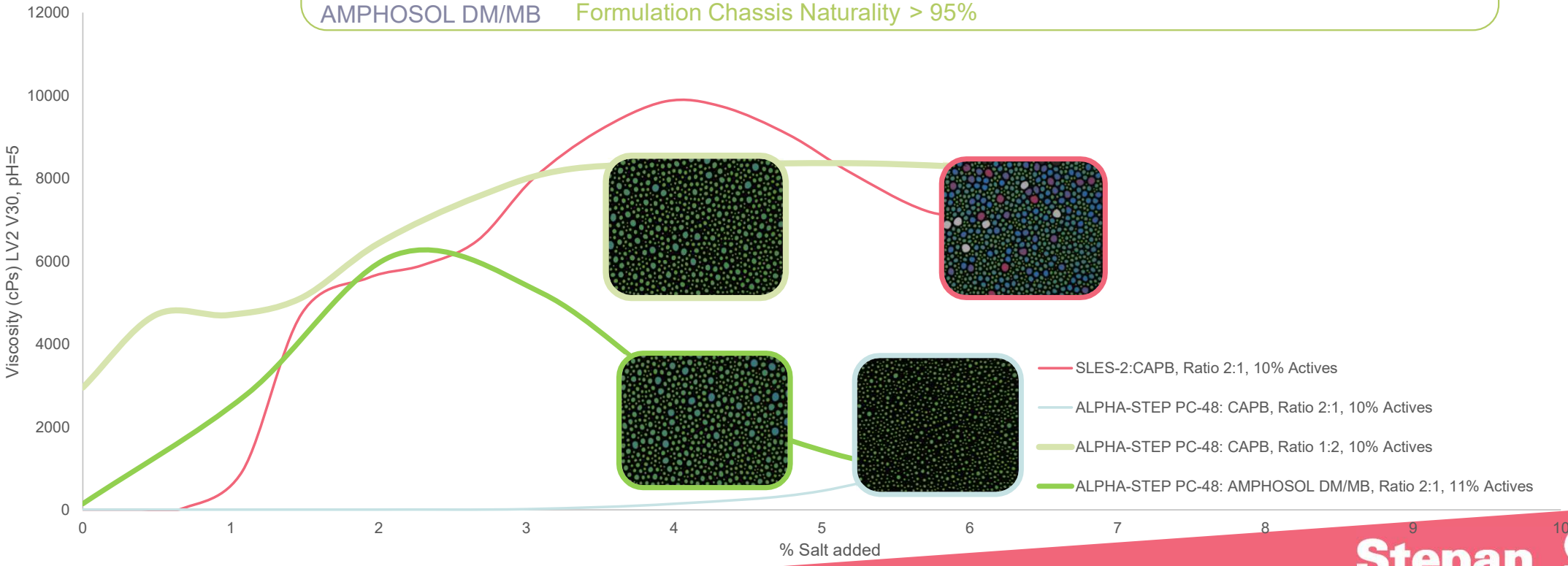
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Sulfonate Thickening Hacks

Achieve similar performance to benchmark SLES-2/CAPB **while improving sensory and naturality**

#5

ALPHA-STEP PC-48 CAPB **or** ALPHA-STEP PC-48 AMPHOSOL DM/MB	Equal viscosity by using ALPHA-STEP PC-48 and AMPHOSOL DM/MB at a standard surfactant ratio. Minimize salt addition, when ALPHA-STEP PC-48 is combined with CAPB in an inverted surfactants ratio. Foam sensory enhancement – smaller bubble size, creaminess Formulation Chassis Naturality > 95%



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Formulating for
Lubricity, Creamy foam & Mildness

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LATHANOL LAL COARSE/MB

INCI Name	Sodium Lauryl Sulfoacetate
Appearance	White powder
Active	65%
Preservative	None

Cream, Rich Lather
Moisturized after feel
Mild cleaner

BCI=86
Readily biodegradable
No preservative

Easier to solubilize than isethionates
Create low-water products
Achieve water-white formulas



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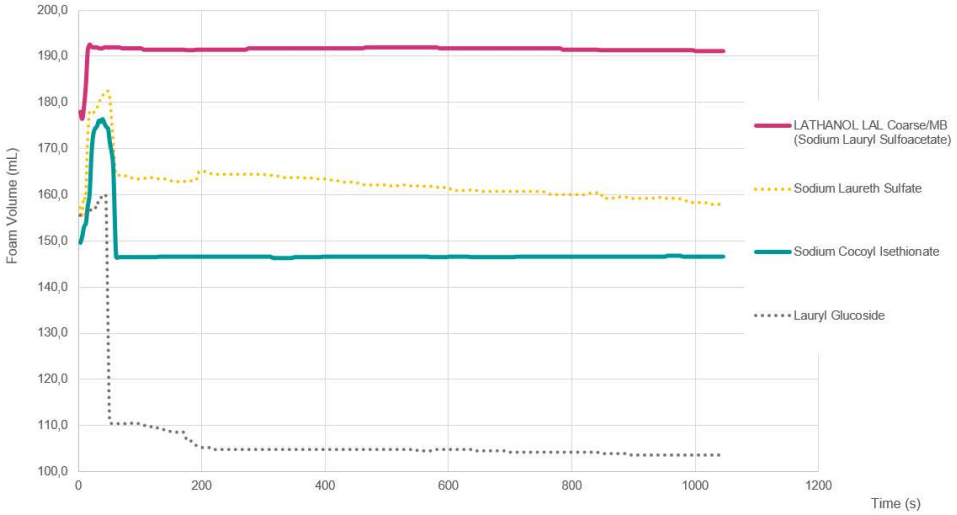
Sulfate-Free Alternatives to SLES

LATHANOL LAL COARSE/MB: Rich Lather, Pampered Feel

Foam Profile

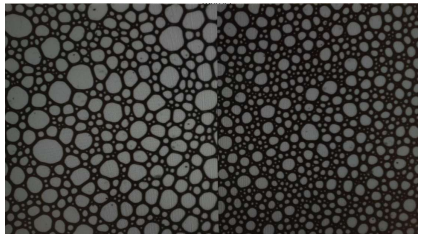
LATHANOL LAL COARSE/MB is a high-foaming co-surfactant for liquid and solid format formulations. It creates more stable foam with rich and creamy attributes compared to SLES-2 and other surfactants (isethionate), thanks to its tighter foam structure*.

Foam Volume Over Time



*See Pixie Dust Stories for LATHANOL LAL COARSE/MB in action
Episode 1: Going Packaging free
Episode 2: Sparkly Bubbli Boo
Episode 3: A Genie in the Beaker

Dynamic Foam Scan Analyser (DFA 100)





0.05% Sodium Laureth Sulfate
In DI Water

0.04% Sodium Laureth Sulfate
0.01% LATHANOL LAL COARSE/MB
In DI Water

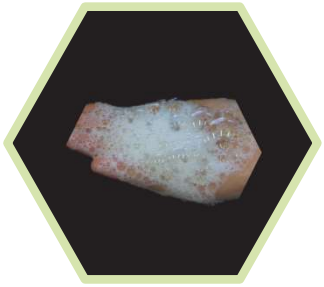
Foam Generation

Reference
SLES-2/CAPB: 12% actives



Good foam volume/Light foam
Big bubbles

Reference
+ 1% LATHANOL LAL COARSE/MB



Creamy foam
Fine and dense bubbles
Pleasant skin feel



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Formulating for Simplicity

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STEPAN-MILD PCL-BA

INCI Name	Sodium Lauryl Sulfoacetate (and) Sodium Methyl 2-Sulfolaurate (and) Disodium 2-Sulfolaurate
Appearance	Clear Yellow Liquid
Active	23%
Preservative	Benzoic Acid

Creamy, rich lather
Moisturized skin after-feel
Rinses cleanly and easily

90% Bio-based
Vegetable feedstock
Biodegradable

Easy handling and formulating
Incorporate high oil loads
Achieve water white formulas
Globally-approved preservative

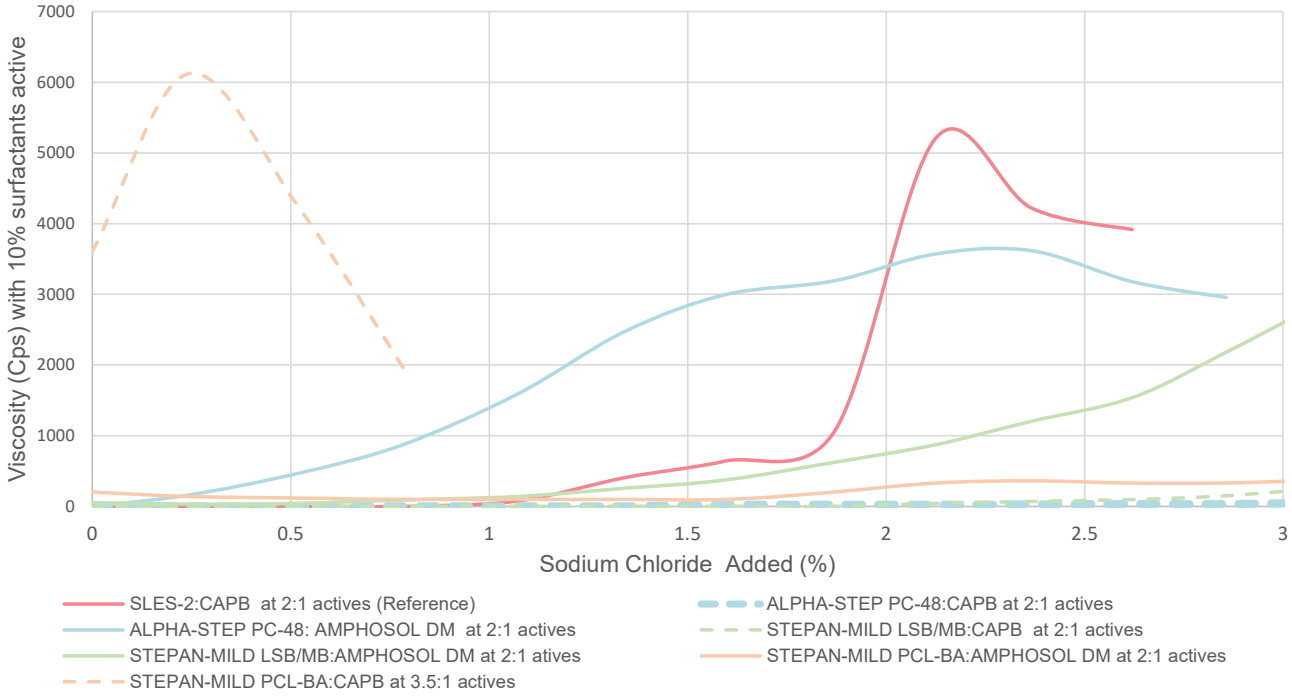


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Sulfate-Free Alternatives to SLES
Specialty Primary Foamers

Better Viscosity Building than CAPB

Both AMPHOSOL DM/MB and AMPHOSOL CDB-HP-E can provide excellent thickening with a majority of sulfate-free anionic surfactants, better than the standard secondary CAPB. In the example below, STEPAN-MILD LSB/MB and ALPHA-STEP PC-48 fail to build viscosity when combined with CAPB. Viscosity building is significantly improved with AMPHOSOL DM/MB. The best secondary surfactant is CAPB for STEPAN-MILD PCL-BA.



Foam Profiles

STEPAN-MILD LSB/MB and STEPAN MILD PCL-BA create rich and creamy foam, while ALPHA-STEP PC-48 creates a light and airy foam.



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Formulation Inspiration

Ether Sulfate-Free Option with BIO-TERGE AS-40 HASB

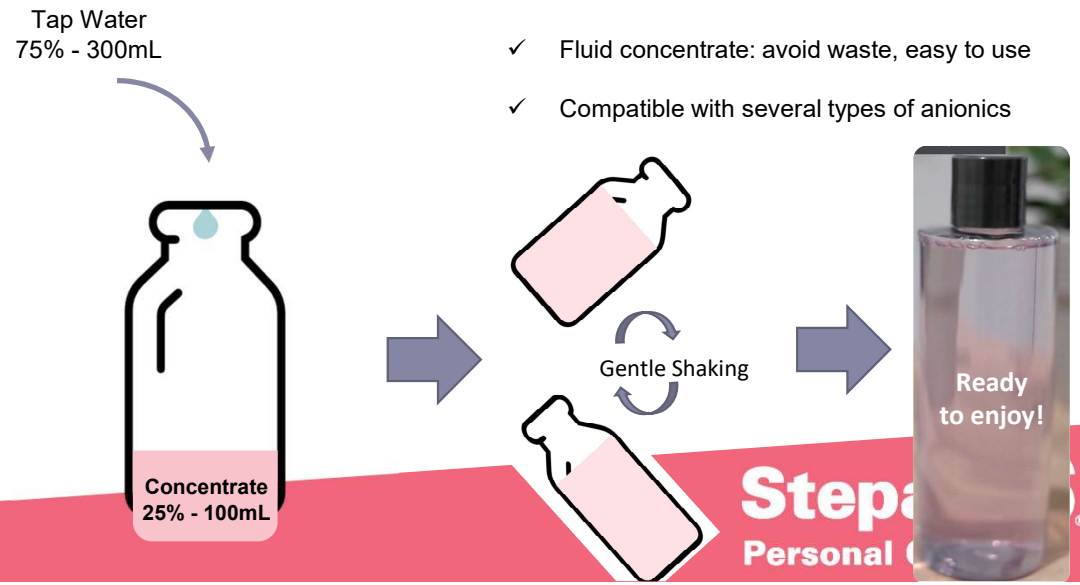
AOS-Based Concentrated Shower Gel

Formulation No 1443

This formulation is an alpha olefin sulfonate (AOS)-based, concentrated shower gel. Dilute the formulation with tap water at a 1:3 ratio. Using 100 mL of concentrate will create a 400 mL shower gel. Recycle an emptied shower gel bottle or use a glass bottle for an eco-conscious package option. This formulation has a Natural Origin Content (NOC) of 0.86 and an NOC of 0.96 when diluted at a 1:3 ratio as outlined in ISO 16128.

Watch the video

Ingredients	% by weight
Deionized Water	q.s. to 100.0
STEPAN-MILD PCL-BA *Sodium Lauryl Sulfoacetate and Sodium Methyl 2-Sulfolaurate (and) Disodium 2-Sulfolaurate*	**37.3**
AMPHOSOL DM/MB *Coco Betaine*	**37.3**
BIO-TERGE AS-40 HASB *Sodium C14-16 Olefin Sulfonate*	**8.6**
Glycerin	1.9
Propylene Carbonate	2.0
Citric Acid (50% Solution)	q.s
Fragrance, preservative, dye	q.s



- ✓ Fluid concentrate: avoid waste, easy to use
- ✓ Compatible with several types of anionics

Formulation Inspiration

Gentle Cleansing Body Wash

Gentle Cleansing Body Wash

Formulation No 1363

If you need some moisturization in your daily body wash, use this sulfate-free cleanser formulated with STEPAN-MILD® L3, a natural moisturizing agent, which replaces synthetic silicones. Squeeze it onto a shower pouf or directly to your hands and enjoy the rich, creamy lather.

Ingredients	% by weight
Deionized Water	q.s. to 100.00
MAPROSYL® 30-B	17.00
STEPAN-MILD® PCL-BA	29.00
AMPHOSOL® CDB-HP	11.00
Glucamate DOE-120 (The Lubrizol Corporation) PEG-120 Methyl Glucose Dioleate	1.00
Tween™ 20 (Croda International Plc.) Polysorbate 20	1.00
STEPAN-MILD® L3	0.50
DPG (Spectrum Chemical Mfg. Corp.) Dipropylene Glycol	1.00
Fragrance	0.50
Microcare® SBB (THOR Specialties, Inc.) Benzyl Alcohol,	0.25
Benzoic Acid & Sorbic Acid	q.s.
Citric Acid (50% Solution)	q.s.
Sodium Chloride	
Total	**100.00**

Formulation Inspiration

Shower Caresse

Shower Caresse

Formulation No 1487

Give skin a morning shower caresse with this mild, naturally derived and sulfate-free shower gel. Enhance foam lather with LATHANOL® LAL COARSE/MB and skin sensory feel with STEPAN-MILD® GCC-FL/MB.

Ingredients	% by weight
Deionized Water	64.30
ALPHA-STEP® PC-48	19.00
LATHANOL® LAL COARSE/MB	1.20
AMPHOSOL® DM/MB	13.00
euxyl™ k 712 (Ashland, Inc.)	1.00
Parfum Soin Intense 20 (Givaudan)	0.20
Dye	q.s.
Citric Acid	0.20
STEPAN-MILD® GCC-FL/MB	1.10
Total	**100.00**

Formulation Inspiration

Silky Foaming Hand Soap

Silky Foaming Hand Soap

Formulation No 1501

This is a sulfate-free, EO-free (ethylene oxide free), and nitrosamine free hand soap based on ALPHA-STEP® PC-48. This sulfate-free and EO-free surfactant enables to produce a light and airy foam. AMPHOSOL® DM/MB enhanced mildness and is an excellent thickening agent for tougher sulfate-free systems and is a COSMOS approved ingredient. STEPAN-MILD® GCC-FL/MB, a naturally derived and COSMOS approved ingredient, is used as foam volume booster and gives a soft and cushioned feel to the skin. This formulation has a low active level at 5.5% total actives, and can be formulated without heat. The water thin viscosity of this formulation is designed for a foaming hand soap dispenser.

Ingredients	% by weight
Deionized Water	83.44
ALPHA-STEP® PC-48	9.40
AMPHOSOL® DM/MB	5.91
euxyl™ k 712 (Ashland, Inc.)	1.00
Silky Touch (Mane SA) Fragrance	0.05
Citric Acid	q.s.
STEPAN-MILD® GCC-FL/MB	0.20
Total	**100.00**

Formulation Inspiration

Powder Milk Foam

Powder Milk Foam

Formulation No 1520

Soap powder for hand washing that provides cleaning and is very easy to carry in your handbag and backpack, in addition to not containing water in the formulation, which helps to preserve water.

Ingredients	% by weight
Dry-Flo® Pure (Nouryon) Aluminum Starch Octenylsuccinate	q.s.p to 100%
BIO-TERGE® AS-90 BEADS	21.00
LATHANOL® LAL POWDER	2.00
NEOBEE® M-5 COSMETIC	2.00
STEPANQUAT® Soleil	1.00
0623-000167 Leite Ninho (Dierberger)	2.00
Dye	q.s.
Total	**100.00**

Formulation Inspiration

Choco Delight: Powder Shampoo

Choco Delight: Powder Shampoo

Formulation No 1488

Bored with the same cleansing routine? Try this fun powder shampoo and delight your senses with a smooth foam, soft feel and delicious chocolate aroma. This waterless formulation is sulfate, silicone and ethylene oxide-free. Use it in its powder form or mix it with water to create a smooth mousse.

Ingredients	% by weight
Citric Acid	18.00
Sodium Bicarbonate	17.00
Corn Starch (El Molino Verde) Zea Mays (Corn Starch)	19.50
LATHANOL® LAL Powder	32.00
BIO-TERGE® AS-90 Coarse	10.00
Cosmetic Brown Iron Oxide (Be Organic) CI 77499	0.50
STEPAN-MILD® L3	2.00
Argan Oil (Indchemie) Argania Spinosa Kernel Oil	0.50
Venus (Firmenich)	0.50
Total	**100.00**

Formulation Inspiration

Sparkli-Bubbli-boo : Powdered Hair & Body wash

Sparkli-Bubbli-boo : Powdered Hair & Body wash

Formulation No 1374

Bored with the same cleansing routine? Try this fun powder shampoo and delight your senses with a smooth foam, soft feel and delicious chocolate aroma. This waterless formulation is sulfate, silicone and ethylene oxide-free. Use it in its powder form or mix it with water to create a smooth mousse.

Ingredients	% by Weight
Nativacare 8600 (Ingredion Incorporated) Rice Starch	35.7
STEPANOL® WA-100	9.0
LATHANOL® LAL COARSE/MB	28.0
Citric Acid	18.0
Sodium Carbonate	9.0
Fragrance	0.3
Total	**100.0**

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